SAVNav: Socially-Aware Audio Visual Navigation

1st Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

***Abstract*—This document is a model and instructions for LaTeX. This and the IEEEtran.cls file define the components of your paper [title, text, heads, etc.]. *CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.**

***Index Terms*—component, formatting, style, styling, insert**

I. METHODOLOGY

A. SYSTEM OVERVIEW & PROBLEM FORMULATION

We consider a mobile robot operating in a dynamic, semi-structured indoor environment populated by human agents. The robot is equipped with an RGB-D camera and a 4-channel microphone array.

Coordinate Systems: We define the World Frame $\mathcal{W}$ with the origin at the scene center and the Ego Frame $\mathcal{E}$ attached to the robot's geometric center. Let $\mathbf{x}_t = [x, y, z, \psi]^T$ denote the robot's pose in $\mathcal{W}$ at time t.

Problem Definition: Given a natural language instruction $\mathcal{I}$ (e.g., *"Find the person talking"*), the objective is to generate a sequence of control actions $a_t \in \mathcal{A}$ to navigate towards the target location $\mathbf{g}^*$ while minimizing the traversal time and ensuring social compliance. This requires handling two types of entities: Observable Entities (within the camera's Field-of-View, FoV) and Occluded Acoustic Sources (Non-Line-of-Sight, NLOS).

B. MULTI-MODAL PERCEPTION WITH UNCERTAINTY

Our perception stack processes visual and auditory streams asynchronously to extract semantic and spatial information, explicitly modeling the uncertainty inherent in each modality.

1) Visual Perception: The visual module runs at camera frame rate ($\sim$ 20Hz). We employ YOLOv11 [1] for 2D object detection and segmentation. For each detected bounding box, we compute the 3D centroid $\mathbf{p}_{vis}^{\mathcal{E}}$ in the Ego Frame using depth data.
To account for sensor noise and partial occlusions, we model the positional uncertainty σ_p as:

$$\sigma_p = k_d \cdot z^2 + k_{occ} \cdot \left(1 - \frac{N_{mask}}{N_{box}}\right) \tag{1}$$

where z is the depth, k_d captures the depth sensor's quadratic error characteristics, and the second term penalizes occlusion based on the mask-to-box ratio with coefficient k_{occ}.
We extract semantic features $\mathbf{f}_{vis}$ using ImageBind [2] and appearance features $\mathbf{f}_{reid}$ for temporal tracking.

Identify applicable funding agency here. If none, delete this.

2) Audio Perception: The audio module processes 4-channel ambisonic signals in a sliding window (5s duration, 0.5s stride). We utilize Embed-ACCDOA [3] for Sound Event Localization and Detection (SELD).
For each frame, the model predicts the Direction-of-Arrival (DoA) vector $\mathbf{v}_{doa} \in \mathbb{R}^3$ and a semantic label. The angular uncertainty σ_{ang} is dynamically scaled by the detection confidence c:

$$\sigma_{ang} = \sigma_{base} \cdot (1 + k_{unc} \cdot (1 - c)) \tag{2}$$

where σ_{base} is the empirical error margin of the SELD model. Unlike vision, audio lacks direct distance information; thus, we model the source position as a probability cone $\mathcal{C}_{audio}(\mathbf{x}_t, \mathbf{v}_{doa}, \sigma_{ang})$ originating from the robot.

C. SAVMAP: SPATIO-TEMPORAL MAPPING

We introduce SAVMap, a layered grid map (0.05m resolution) that integrates static semantics with dynamic audio-visual entities to support socially-aware navigation.

1) Entity Tracking and Association: We maintain a list of tracked entities. For visual observations, we employ a Hungarian matching algorithm based on a hybrid cost matrix C_{ij}:

$$C_{ij} = w_{pos} \cdot \|\mathbf{p}_i - \mathbf{p}_j\| + w_{emb} \cdot (1 - \cos(\mathbf{f}_i, \mathbf{f}_j)) \tag{3}$$

For audio observations, we perform cross-modal association. If an audio source's DoA aligns with a visual entity within the angular threshold σ_{ang} and shares semantic consistency (e.g., "speech" vs. "person"), the audio event is anchored to the visual target.

2) Topology-Aware Hallucinated Momentum: A key contribution of our work is handling unanchored audio sources (i.e., heard but not seen). To anticipate NLOS risks (e.g., a person approaching from around a corner), we propose a Hallucinated Momentum Injection mechanism:

1. Ray-Casting: We project a ray along the unanchored DoA until it intersects with the static map boundaries.
2. Topological Adsorption: The intersection point is snapped to the nearest topological node (e.g., a doorway or corner), denoted as the hallucinated position $\mathbf{p}_{hall}$.
3. Virtual Velocity: We assign a virtual velocity vector $\mathbf{v}_{virt}$ to this entity, pointing directly towards the robot, simulating a worst-case collision scenario.

3) Social Cost Layer: We construct a unified cost map M_{cost} to guide the planner.

1. Visible Humans: For tracked visual entities, we generate a Gaussian cost field. If the human is moving, the Gaussian is anisotropic, elongated along their velocity vector.
2. NLOS Audio Sources: For hallucinated entities, we generate a Lance-shaped Risk Field aligned with $\mathbf{v}_{virt}$ to penalize the path directly in front of the hidden source:

$$J_{audio}(\mathbf{x}) = A \cdot \exp\left(-\left(\frac{d_{long}^2}{2\sigma_{long}^2} + \frac{d_{lat}^2}{2\sigma_{lat}^2}\right)\right) \quad (4)$$

where d_{long} and d_{lat} are longitudinal and lateral distances to the source in the frame aligned with $\mathbf{v}_{virt}$. By setting $\sigma_{long} \gg \sigma_{lat}$, this cost forces the planner to take wide turns at corners ("cutting the corner wide"), effectively implementing a "Hearing before Seeing" behavior.

D. SAVNAV: SOCIALLY-AWARE PLANNING

To handle the ambiguity of audio data, we design a dual-mode planner that switches between Precision Navigation and Active Exploration.

1) State Arbitration: The system state is determined by the target entity's status:

- State A (Precision Nav): Triggered when the target is visually anchored or statically defined (e.g., "Door").
- State B (Active Exploration): Triggered when the target is only detected via audio (Unanchored).

2) State A: Risk-Aware Path Generation: In this state, the goal $\mathbf{g}^*$ is deterministic. We employ the Fast Marching Method (FMM) to solve the Eikonal equation over the speed field $S(\mathbf{x}) = S_{max} \cdot (1 - M_{cost}(\mathbf{x}))$. The gradient descent on the resulting arrival time map $T(\mathbf{x})$ yields a smooth, global path that naturally skirts high-social-cost regions.

3) State B: Information-Theoretic Active Exploration: When the target position is uncertain (audio-only), the robot must actively reduce entropy. We define the Region of Interest (ROI) Ω_{ROI} as the intersection of the audio uncertainty cone $\mathcal{C}_{audio}$ and the navigable space.

We select the Next-Best-View (NBV) pose $\mathbf{p}_{nbv}$ by maximizing an Information-Theoretic Utility function $U(\mathbf{p})$:

$$U(\mathbf{p}) = \underbrace{\text{Area}(\text{FOV}(\mathbf{p}) \cap \Omega_{ROI} \setminus \mathcal{V}_{hist})}_{\mathcal{I}_{gain}} \cdot \underbrace{\exp(-\lambda \cdot \|\mathbf{x}_t - \mathbf{p}\|)}_{\mathcal{C}_{cost}} \quad (5)$$

where $\mathcal{I}_{gain}$ represents the expected volumetric information gain (unobserved ROI area visible from $\mathbf{p}$), and $\mathcal{C}_{cost}$ penalizes travel distance. $\mathcal{V}_{hist}$ tracks previously observed regions to prevent redundant exploration.

Once the robot reaches $\mathbf{p}_{nbv}$ or the target enters the FoV during transit, the system transitions to State A for final approach.

4) Reactive Control: The planned path is executed by a local controller. To ensure safety, an Emergency Brake is triggered if the Time-To-Collision (TTC) with any dynamic obstacle (visual or hallucinated) drops below 1.0s.

REFERENCES

[1] R. Khanam and M. Hussain, "YOLOv11: An overview of the key architectural enhancements," 2024.

[2] R. Girdhar, A. El-Nouby, Z. Liu, M. Singh, K. V. Alwala, A. Joulin, and I. Misra, "ImageBind one embedding space to bind them all," in *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. Vancouver, BC, Canada: IEEE, Jun. 2023, pp. 15 180–15 190.

[3] K. Shimada, K. Uchida, Y. Koyama, T. Shibuya, S. Takahashi, Y. Mitsufuji, and T. Kawahara, "Open-vocabulary sound event localization and detection with joint learning of CLAP embedding and activity-coupled cartesian DOA vector," *IEEE Transactions on Audio, Speech, and Language Processing*, vol. 33, pp. 2946–2960, 2025.